

CSE565-Software-Verif-Validation-Test

Developing Test Cases with DOE Tool

Tool: PICT

Test Case	Type of Phone	Authentication	Connectivity	Memory	Battery Level
TC1	iPhone 13	Fingerprint	5G Edge	128 GB	40 ~ 59%
TC2	iPhone 14	Text Password	Wireless	512 GB	80 ~ 100%
TC3	Huawei Mate	Face Recognition	4G LTE	256 GB	40 ~ 59%
TC4	Google Pixel 7	Fingerprint	3G	1 TB	< 20%
TC5	iPhone 13	Text Password	4G LTE	1 TB	20 ~ 39%
TC6	Google Pixel 7	Face Recognition	Wireless	128 GB	20 ~ 39%
TC7	Galaxy Z	Text Password	3G	256 GB	80 ~ 100%
TC8	Google Pixel 7	Face Recognition	5G Edge	512 GB	60 ~ 79%
TC9	Huawei Mate	Text Password	3G	128 GB	60 ~ 79%
TC10	iPhone 13	Fingerprint	4G LTE	512 GB	80 ~ 100%
TC11	iPhone 13	Fingerprint	Wireless	256 GB	60 ~ 79%
TC12	Google Pixel 7	Text Password	Wireless	1 TB	40 ~ 59%
TC13	Huawei Mate	Fingerprint	3G	512 GB	20 ~ 39%
TC14	Galaxy Z	Face Recognition	5G Edge	1 TB	60 ~ 79%
TC15	iPhone 14	Face Recognition	5G Edge	256 GB	20 ~ 39%
TC16	Google Pixel 7	Face Recognition	4G LTE	128 GB	80 ~ 100%
TC17	iPhone 14	Text Password	4G LTE	128 GB	< 20%
TC18	Galaxy Z	Fingerprint	Wireless	512 GB	< 20%
TC19	Galaxy Z	Face Recognition	3G	512 GB	40 ~ 59%
TC20	Huawei Mate	Text Password	5G Edge	1 TB	< 20%
TC21	iPhone 14	Fingerprint	4G LTE	1 TB	60 ~ 79%
TC22	Huawei Mate	Face Recognition	Wireless	1 TB	80 ~ 100%
TC23	iPhone 13	Face Recognition	3G	256 GB	< 20%
TC24	iPhone 14	Text Password	3G	256 GB	40 ~ 59%
TC25	Galaxy Z	Text Password	4G LTE	128 GB	20 ~ 39%

Test Case	Type of Phone	Authentication	Connectivity	Memory	Battery Level
TC26	Google Pixel 7	Text Password	5G Edge	256 GB	80 ~ 100%

Developing Test Cases with Gen AI Tool

Tool: ChatGPT

Test Case	Type of Phone	Authentication	Connectivity	Memory	Battery Level
TC1	iPhone 14	Fingerprint	Wireless	128 GB	<20%
TC2	iPhone 13	Face Recognition	3G	256 GB	20–39%
TC3	Galaxy Z	Text Password	4G LTE	512 GB	40–59%
TC4	Huawei Mate	Fingerprint	5G Edge	1 TB	60–79%
TC5	Google Pixel 7	Face Recognition	Wireless	512 GB	80–100%
TC6	iPhone 13	Text Password	4G LTE	128 GB	60–79%
TC7	Galaxy Z	Fingerprint	3G	1 TB	<20%
TC8	Huawei Mate	Face Recognition	Wireless	256 GB	40–59%
TC9	Google Pixel 7	Text Password	5G Edge	128 GB	20–39%
TC10	iPhone 14	Face Recognition	4G LTE	1 TB	80–100%
TC11	Galaxy Z	Text Password	Wireless	256 GB	60–79%
TC12	Huawei Mate	Text Password	3G	512 GB	20–39%
TC13	Google Pixel 7	Fingerprint	5G Edge	256 GB	<20%
TC14	iPhone 14	Text Password	3G	512 GB	40–59%
TC15	iPhone 13	Fingerprint	5G Edge	128 GB	80–100%

Comparison between Tools

Criterion	DOE Tool (Task 1)	Generative AI Tool (Task 2)
Number of Test Cases	26 cases	15 cases
Coverage Guarantee	Ensures complete pairwise coverage for all parameter pairs	Coverage may be incomplete; depends on AI prompt and random variation
Reliability / Consistency	Deterministic and repeatable; same inputs always yield same set	Non-deterministic; results may vary across prompts or sessions
Ease of Generation	Requires setup (parameter input, DOE configuration)	Very fast; generated instantly with a single prompt

Criterion	DOE Tool (Task 1)	Generative AI Tool (Task 2)
Accuracy and Verification	High accuracy; automatically validated for pairwise completeness	Requires manual verification to ensure all pairs are covered
Flexibility	Rigid parameter setup; harder to include natural-language constraints	Highly flexible; constraints and preferences can be described in plain English
Output Format	Structured table with clear test case order	May require formatting or cleanup for consistency
Learning Curve	Moderate; user must understand DOE concepts	Low; anyone can prompt the AI without specialized knowledge
Best Use Case	Formal software testing requiring guaranteed coverage and reproducibility	Early-stage brainstorming or exploratory test generation
Overall Assessment	Most reliable and systematic approach	Fastest and most adaptable but less rigorous

DOE Tool Assessment (PICT)

Features and Functionalities

The Microsoft PICT tool offers a robust and efficient solution for generating pairwise test cases based on combinatorial design. Its primary functionality lies in automatically creating a minimal set of test cases that ensures every pair of parameter values appears at least once. PICT also supports advanced features such as constraint handling, which allows users to exclude invalid or impossible combinations, and weighted parameters, which help prioritize certain values over others.

Scope Covered by the Tool

The scope of PICT covers a wide range of combinatorial testing needs, from simple pairwise coverage to more complex n-wise configurations. In this project, PICT was applied to five parameters and each with multiple possible values. The full factorial design would have required 1,200 test cases, but PICT efficiently reduced this to only 26 test cases while maintaining complete pairwise coverage. This demonstrates the tool’s capability to significantly reduce testing effort without compromising test completeness. The results covered all logical pairwise interactions, making it suitable for both functional and configuration-based testing scenarios.

Performance

PICT performed exceptionally well in terms of speed, accuracy, and consistency. The tool generated the full set of pairwise test cases almost instantly after providing the model file, showcasing its computational efficiency. It is deterministic, meaning that running the same model multiple times yields identical results, which is an essential feature for reproducible and traceable test design. PICT’s algorithm optimizes test generation by minimizing redundancy and ensuring even distribution of parameter combinations. Overall, its performance was highly reliable and consistent, producing results that align with the mathematical guarantees of the pairwise DOE technique.

Ease of Use

While PICT is lightweight and fast, its and text-based model setup can pose a challenge for beginners unfamiliar with DOE concepts. Users must manually define all parameters and their possible values, which requires attention to syntax and formatting. However, once the initial setup is complete, generating test cases is straightforward. The output is in a clean tab-delimited format that can be easily imported into spreadsheets or documentation. Thus, while the learning curve is moderate, the tool is highly efficient and repeatable once mastered, making it a dependable choice for systematic test design.

Gen AI Tools Assessment (ChatGPT)

Instead of manually defining parameters in a structured input file as required by PICT, I was able to describe the problem in natural language and asking ChatGPT to create a minimal set of pairwise test cases. The AI quickly produced a well-structured table containing 15 test cases that covered most combinations with minimal overlap. Crafting the prompt required some iteration; the initial output contained a few redundant or missing pairs, but refining the prompt with clearer instructions (such as “ensure every pair of parameter values appears at least once”) improved the results significantly. The generated test cases were easy to read and interpret, and formatting them into a spreadsheet or report required minimal post-processing. Overall, ChatGPT offered a fast, accessible, and user-friendly way to create test designs without needing prior knowledge of DOE tools.

The use of generative AI in software testing introduces a new level of speed, flexibility, and creativity to the test design process. While traditional DOE tools like PICT guarantee mathematical completeness, ChatGPT enables testers to explore and adjust test case designs rapidly through natural-language interaction. This is particularly useful in early development stages when requirements evolve frequently, or when human testers need quick drafts of test combinations for brainstorming or exploratory testing. Although AI-generated outputs may require manual verification to confirm full pairwise coverage, the productivity gains and reduced learning curve are substantial. In summary, ChatGPT demonstrates how generative AI can complement conventional DOE tools—offering agility and accessibility while accelerating the overall software testing workflow.